

Information Technology University of the Punjab
SE460T Cloud Computing with Big Data Applications – Fall 2025
Quiz 2 Section B – November 5, 2025

Name: _____

Roll No.: _____

Time allowed: 30 minutes

Maximum Marks: 25

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1. A company is launching a Linux instance that will be used for batch processing. They want to select an instance type that is best suited for the workload. Which instance type should they choose when launching the instance? [1]
 - a. Compute optimized instance type
 - b. General purpose instance type
 - c. Accelerated computing instance type
 - d. Memory optimized instance type
 2. A company is launching a Linux instance that will be used to host a high-performing Redis database. They want to select an instance type that is best suited for the workload. Which instance type should they choose when they launch the instance? [1]
 - a. High performance optimized instance type
 - b. Accelerated computing instance type
 - c. Compute optimized instance type
 - d. Memory optimized instance type
 3. A company needs to launch 300 Linux EC2 instances. All 300 Linux instances need to have access to shared files. Which storage solution should they choose to share files among the 300 Linux instances? [1]
 - a. Amazon Elastic Block Store (Amazon EBS)
 - b. Amazon FSx for Windows File Server
 - c. Amazon Elastic File System (Amazon EFS)
 - d. Amazon instance store
 4. A company need to launch a Linux instance that will host a database. The instance needs to stop on occasion to perform maintenance tasks. Which storage solution should they choose for the data? [1]
 - a. Amazon Elastic File System (Amazon EFS)
 - b. Amazon Elastic Block Store (Amazon EBS)
 - c. Amazon FSx for Windows File Server
 - d. Amazon instance store
 5. You need to attach the same data volume to multiple EC2 instances across Availability Zones. Which service should you use? [1]
 - a. Amazon EBS
 - b. Amazon EFS
 - c. Instance store
 - d. Amazon S3
 6. Which EC2 instance type family is optimized for compute-intensive workloads such as gaming or scientific modeling? [1]
 - a. R family
 - b. T family
 - c. C family
 - d. M family
 7. What happens to data on the root EBS volume of an EC2 instance when the instance is terminated, by default? [1]
 - a. It is preserved
 - b. It is deleted
 - c. It is moved to S3
 - d. It is detached and retained as an orphan volume
 8. You are designing a compute layer that requires fast, temporary storage which is lost on stop/terminate. Which storage type fits this need? [1]
 - a. Amazon EFS
 - b. Instance store
 - c. Amazon EBS
 - d. Amazon S3
 9. Which of the following is NOT a database engine supported by Amazon RDS? [1]
 - a. MySQL
 - b. PostgreSQL
 - c. Cassandra
 - d. MariaDB
 10. What is the primary benefit of using Amazon RDS over deploying a database on EC2? [1]
 - a. RDS offers complete control of the operating system
 - b. RDS allows SSH access to the DB instance
 - c. RDS supports containerized deployment
 - d. RDS automatically manages backups, patching, and failover
 11. You are designing an RDS solution that must withstand AZ failure automatically. Which configuration should you choose? [1]
 - a. Single-AZ deployment
 - b. Read replica
 - c. Aurora Serverless
 - d. Multi-AZ deployment
 12. In an RDS Multi-AZ, the standby instance: [1]
 - a. Is automatically promoted during failover
 - b. Is used for read scaling
 - c. Can be used for backups
 - d. Can be manually connected for read queries
 13. In DynamoDB, which type of index allows queries on attributes other than the primary key and can have a different partition/sort key schema? [1]
 - a. Local Secondary Index (LSI)
 - b. Global Secondary Index (GSI)
 - c. Primary Index
 - d. Secondary Table Index

14. Which of the following cannot be changed after a DynamoDB table is created? [1]

- a. Partition key
- b. Read/write capacity
- c. Table name
- d. Encryption setting

15. Which DynamoDB feature allows you to track and respond to item-level changes in real time? [1]

- a. DAX
- b. Streams
- c. TTL
- d. Global Tables

16. In the RDS Challenge Lab, what are the key differences between the architecture at the start of the lab and the architecture after completion of the lab? [3]

The architecture at the start of the lab has MariaDB engine running on the same EC2 instance that is running the web server. The architecture at the end of the lab has the same MariaDB database lying on an RDS for MariaDB instance. The web application in the EC2 instance now accesses the duplicate Database on the RDS instance.

17. After the database has been migrated to RDS for MariaDB, how does the CafeApplication running on the EC2 instance access this migrated database? [3]

The CaféApplication accesses the migrated database by using the updated secrets in the Secrets Manager, namely, the RDS endpoint, new username, and new password.

18. After duplicating the AMI and launching it on an EC2 instance in the Oregon region, do we make any changes to the new AMI? How do we change the secrets manager parameters in the new Oregon region? [4]

No, we do not make any changes to the new AMI. We change the setapp_parameter.sh file in the previous EC2 instance with updates about region, EC2 URL, etc. This script is then run which configures the Secrets Manager secrets in the region of our new EC2 instance.